TECHNICAL PROJECT PROPOSAL

Predictive Claims Valuation in Workers' Compensation

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1. Executive Business Context & Problem Framing

Worker's Compensation insurance requires reserves to be set aside as soon as an injury is reported. Initial case estimates (`InitialCaseEstimate`) set by claims adjusters are often subjective and subject to severe revision over time as medical treatments progress.

- **Business Objective:** Build an automated, data-driven ML pipeline to accurately predict final settlement cost (`UltimateIncurredClaimCost`) early in the claim lifecycle.
- **Operational Impact:** Enable early triage of severe claims (Top 10% highest cost) to assign experienced adjusters, optimize financial reserve allocation, and mitigate severe tail-risk losses.

2. Dataset Architecture & Leak-Proof Preprocessing

The dataset comprises $100{,}000$ historical claim records combining demographic attributes (`Age`, `Gender`, `MaritalStatus`), financial status (`WeeklyPay`, `HoursWorkedPerWeek`, `DaysWorkedPerWeek`), family dependents (`DependentChildren`, `DependentsOther`), dates, and unstructured text (`ClaimDescription`).

- **Feature Engineering:** Domain feature `ReportDelayDays` calculated strictly as $(\text{DateReported} - \text{DateTimeOfAccident})/86{,}400$ s (converting seconds to days).
- **Strict Leak-Proof Protocol:** Split into 80% Train ($80{,}000$ rows) and 20% Test ($20{,}000$ rows) *prior* to fitting transformers. Scikit-learn `Pipeline` and `ColumnTransformer` wrap `RobustScaler` for numerical features and `OneHotEncoder` for categorical features.

3. Target Variable & Evaluation Benchmark Metrics

The target `UltimateIncurredClaimCost` exhibits a heavy-tailed, highly right-skewed distribution. Models are trained on $\log(1+y)$ with upper-bound prediction clipping at $\max(\log(1+y_{\text{train}}))$ to prevent exponential prediction explosions upon applying $\exp(\cdot)-1$.

- **Primary Regression Metrics:** Root Mean Squared Error (RMSE \$), Mean Absolute Error (MAE \$), Mean Absolute Percentage Error (MAPE %), and R^2 Score.
- **Secondary Triage Metrics:** ROC-AUC and Precision@Top-10% for high-cost claim binary detection (threshold at 90th percentile, $\approx \$12{,}913.88$).

4. Baseline Evaluation & Empirical Benchmark Results

The benchmark was executed on the full $100{,}000$-record dataset (20% held-out test set). Results are summarized below:

Model Family	RMSE ($)	MAE ($)	MAPE (%)	R^2 Score	Fit Time (s)
`dummy_mean_baseline`	50,442.06	11,189.54	313.25	-0.0460	0.072
`linear_regression`	8,992.77	3,802.13	90.83	-0.2753	0.088
`ridge_regression`	8,937.65	3,793.88	90.81	-0.2597	0.076
`lasso_regression`	8,357.92	4,128.06	133.37	-0.1016	0.082
`gradient_boosting`	7,087.54	3,331.90	80.83	+0.2079	0.686
`lightgbm`	6,935.99	**3,301.41**	81.00	+0.2414	1.453
`random_forest`	**6,891.03**	3,307.70	**80.38**	**+0.2512**	0.224

5. Key Technical Findings & Methodological Validation

- **20% Error Reduction:** Non-linear tree ensembles (Random Forest, LightGBM) reduce MAE from $\$4{,}128.06$ down to $\mathbf{\$3{,}301.41}$, proving strong capability to capture complex tabular interactions.
- **Positive R^2 Variance Explained:** R^2 rises from negative values on linear baselines to $\mathbf{+0.2512}$ with Random Forest.
- **Severe Claims Triage:** Baseline achieves ROC-AUC ≈ 0.907 for identifying Top 10% claims, setting a solid classification benchmark.

6. Project Roadmap & Upcoming Milestones (Weeks 10–12)

- **Week 10 (NLP Integration):** Extract text features from `ClaimDescription` using TF-IDF n-grams and fine-tune a pre-trained Transformer (`DistilBERT`) to capture injury severity semantics.
- **Week 11 (Multimodal Fusion & Optimization):** Combine tabular features with text embeddings into a unified LightGBM / Neural Architecture.
- **Week 12 (Interpretability & Final Report):** Apply SHAP (SHapley Additive exPlanations) for global feature importance and local claim risk explanations.